

Noise and Hardness

Distributions and the LWE Family
Lecture 6 of 6 — Foundations

SEC 595 — Encrypted Computing

- 1 The Distributions We Actually Use
- 2 The LWE Family

1.1 Uniform on R_q

Write $a \leftarrow R_q$ for a ring element whose n coefficients are drawn independently and uniformly from $\mathbb{Z}_q$.

If a is uniform and x is any independent ring element, then $a + x$ is uniform: on each coefficient $a \mapsto a + x$ permutes $\mathbb{Z}_q$.

Uniform plus anything is uniform.

ML-KEM and ML-DSA expand a , or the matrix $\mathbf{A}$, from a 32-byte seed with SHAKE-128 — so the public key carries the seed, not the coefficients.

1.2 The ternary key distribution R_2

The secret key is drawn from R_2 : each coefficient in $\{-1, 0, 1\}$, either uniformly or with a prescribed Hamming weight h .

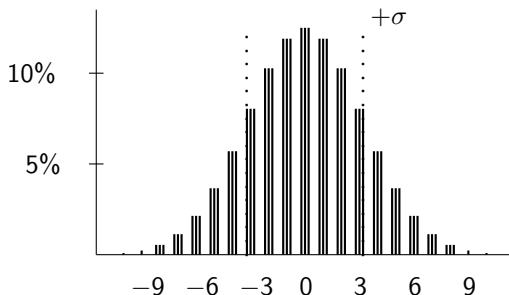
Then $\|SK\|_\infty = 1$, so SK contributes a factor 1 rather than $\beta \approx 19$ wherever it appears in a noise bound.

A sparse secret with $h \ll n$ cheapens the hybrid dual-lattice attacks of Lecture 5, so deployments using one take $h = 128$ or 192 .

Uniform ternary is the default of the Homomorphic Encryption Standard.

1.3 The discrete Gaussian $\mathcal{X}$

The error distribution is a discrete Gaussian over $\mathbb{Z}$, applied to each coefficient; β is an integer a sample exceeds only with negligible probability.



$\Pr[e = x]$ for $D_{\mathbb{Z},8}$, in percent. $\beta = 19$ is off the picture, at six standard deviations.

1.3 The bound β

Definition — discrete Gaussian and the error bound

For $s > 0$ the discrete Gaussian $D_{\mathbb{Z},s}$ assigns to $x \in \mathbb{Z}$ probability proportional to $\exp(-\pi x^2/s^2)$; its standard deviation is $\sigma = s/\sqrt{2\pi}$.

The **error bound** β is an integer such that $e \leftarrow \mathcal{X}$ satisfies $\|e\|_\infty \leq \beta$ with overwhelming probability.

The Homomorphic Encryption Standard takes

$$\mathcal{X} = D_{\mathbb{Z},8}, \quad \mu = 0, \quad \sigma = \frac{8}{\sqrt{2\pi}} \approx 3.1915, \quad \beta = \lceil 6\sigma \rceil = 19.$$

Every norm argument in the scheme lectures uses β , never the distribution.

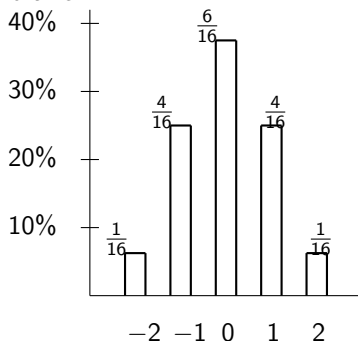
Example — the Standard's numbers

Let $\mathcal{X} = D_{\mathbb{Z},8}$, so $w(x) = \exp(-\pi x^2/64)$ and $\Pr[e = x] = w(x) / \sum_y w(y)$.

- i Compute $\sum_{x \in \mathbb{Z}} w(x)$ to five figures, then $\Pr[e = x]$ for $x = 0, 1, \dots, 6$.
- ii From those, compute $\sigma^2 = \sum_x x^2 \Pr[e = x]$ and check it against $(8/\sqrt{2\pi})^2$.
- iii Compute $\Pr[|e| \leq 9]$, and $\Pr[|e| > 19]$ as a power of two.

1.4 ML-KEM: the centered binomial

FIPS 203 draws each coefficient as $\sum_{i=1}^{\eta} (a_i - b_i)$ from 2η uniform bits: a bit count, a second bit count, a subtraction. No table, no rejection loop, no data-dependent branch.



CBD_2 , used by ML-KEM-768 and ML-KEM-1024.

1.4 What CBD_η is

Proposition

CBD_η is symmetric about 0, supported on $\{-\eta, \dots, \eta\}$, with mean 0 and variance $\eta/2$.

ML-KEM-512 uses $\eta_1 = 3$ for the secret and $\eta_2 = 2$ for the encryption randomness; ML-KEM-768 and ML-KEM-1024 use $\eta_1 = \eta_2 = 2$.

Example — CBD_2 and CBD_3

Each $a_i - b_i$ takes $-1, 0, 1$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$.

- i) Write out the full mass function of CBD_2 as fractions over 16, and say which binomial row it is.
- ii) Verify the mean is 0 and the variance is $\eta/2 = 1$, by direct summation.
- iii) Do the same for CBD_3 : masses over 64, and the variance.

1.5 The four distributions, together

Distribution	Support	Used for
Uniform on R_q	all of $\mathbb{Z}_q$	public a , $\mathbf{A}$
R_2 (ternary)	$\{-1, 0, 1\}$	secret key SK
$\mathcal{X} = D_{\mathbb{Z}, 8}$	$\mathbb{Z}$, bound $\beta = 19$	errors e, e_1, e_2
$\text{CBD}_{\eta}, \eta \in \{2, 3\}$	$\{-\eta, \dots, \eta\}$	ML-KEM secret and error

Two are large and live on R_q ; two are small. A lattice ciphertext is a large uniform part carrying no information plus a small part carrying everything.

2.1 Search and decision LWE

An LWE sample is a linear equation in a secret, deliberately spoiled by a small error.

Search asks for the secret. **Decision** asks only whether the samples have that form or are uniform.

Without the error, n samples give the secret by Gaussian elimination.

2.1 The definition

Definition — LWE, search and decision

Fix n , q , an error distribution $\mathcal{X}$ on $\mathbb{Z}$, and a secret $\mathbf{s} \leftarrow \mathbb{Z}_q^n$. A sample is

$$(\mathbf{a}_i, b_i), \quad \mathbf{a}_i \leftarrow \mathbb{Z}_q^n \text{ uniform}, \quad b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i \bmod q, \quad e_i \leftarrow \mathcal{X}.$$

Search LWE: recover $\mathbf{s}$. **Decision LWE:** distinguish these from uniform on $\mathbb{Z}_q^n \times \mathbb{Z}_q$.

For prime $q = \text{poly}(n)$ the two are equivalent under polynomial-time reductions. By Lecture 5, an LWE sample is a BDD instance on the q -ary lattice $\Lambda_q(\mathbf{A})$.

Example — three samples, one secret

Take $q = 13$, $n = 2$, $\mathbf{s} = (3, 5)$, errors in $\{-1, 0, 1\}$, and the vectors $\mathbf{a} = (1, 2), (4, 7), (5, 2)$ with errors $+1, -1, +1$.

- i) Compute the three samples $(\mathbf{a}_i, b_i)$.
- ii) Search over all 13^2 candidates: which $\mathbf{s}'$ leave every residual in $\{-1, 0, 1\}$? How many survive on only the first two samples?
- iii) With the errors set to 0, recover $\mathbf{s}$ from the first two equations by elimination.

2.2 Ring-LWE

Ring-LWE replaces the matrix by *one* element of $R_q = \mathbb{Z}_q[X]/(X^n + 1)$:

$$(a, b = a \cdot s + e), \quad a \leftarrow R_q \text{ uniform}, \quad e \leftarrow \mathcal{X}^n.$$

One sample is $2n$ elements of $\mathbb{Z}_q$ instead of $n^2 + n$, and the product costs $O(n \log n)$ by the negacyclic NTT of Lecture 3 instead of $O(n^2)$.

The price: the lattice is an *ideal* lattice, not a random q -ary one.

Example — one Ring-LWE sample by hand

Work in $R_{13} = \mathbb{Z}_{13}[X]/(X^2 + 1)$, where $X^2 = -1$. Take

$$s = 2 + 3X, \quad a = 5 + X, \quad e = 1 - X.$$

- i) Compute $a \cdot s$ in R_{13} , reducing with $X^2 = -1$.
- ii) Form the sample (a, b) with $b = a \cdot s + e$.
- iii) How many elements of $\mathbb{Z}_{13}$ does the sample carry, against how many in the secret? State the test that recovers s from (a, b) .

Definition — Module-LWE, with Ring-LWE as the case $k = 1$

Fix R_q of degree n and a rank k ; sample $\mathbf{s} \leftarrow \mathcal{X}^k \subseteq R_q^k$ small. Samples are

$$(\mathbf{A}, \mathbf{b}), \quad \mathbf{A} \leftarrow R_q^{k \times k} \text{ uniform}, \quad \mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}, \quad \mathbf{e} \leftarrow \mathcal{X}^k.$$

At $k = 1$ this is Ring-LWE; at $n = 1$, plain LWE. The lattice has dimension nk .

Security is tuned by changing k alone — n , q , the NTT and every line of arithmetic code stay fixed.

Example — ML-KEM's three parameter sets

All three fix $n = 256$ and $q = 3329$; only k moves, with $k = 2, 3, 4$.

- i) Give the lattice dimension nk for each parameter set.
- ii) For ML-KEM-768, how many ring elements are in $\mathbf{A}$, and how many coefficients of $\mathbb{Z}_{3329}$ is that?
- iii) At $\lceil \log_2 3329 \rceil = 12$ bits each, how many bytes would those coefficients need — and how many does the public key actually send?

Definition — Module-SIS

Given $\mathbf{A} \leftarrow R_q^{k \times \ell}$ uniform and a bound γ , find a nonzero $\mathbf{z} \in R^\ell$ with

$$\mathbf{A}\mathbf{z} = \mathbf{0} \in R_q^k \quad \text{and} \quad \|\mathbf{z}\|_\infty \leq \gamma.$$

Solutions exist for γ large enough by pigeonhole; the problem is *finding* one.

Two distinct valid ML-DSA signatures on one message yield such an element — which is why forgery is hard. MLWE, separately, hides $\mathbf{s}_1$ inside $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$.

Example — a hand-sized MSIS instance

In $R_{13} = \mathbb{Z}_{13}[X]/(X^2 + 1)$ take $k = 1$, $\ell = 2$ and $\mathbf{A} = \begin{pmatrix} 1 & 5 + 2X \end{pmatrix}$.

- i) Write the equation a solution (z_1, z_2) must satisfy.
- ii) Try $z_2 = -2 + X$: compute $(5 + 2X)z_2$ in R_{13} , read off z_1 , and give $\|\mathbf{z}\|_\infty$.
- iii) For which γ is the instance solvable? For which is it trivial?

2.5 Which scheme uses what

Scheme	Assumption	Role
ML-KEM (FIPS 203)	MLWE	IND-CPA of the underlying PKE
ML-DSA (FIPS 204)	MLWE + MSIS	key secrecy + unforgeability
BGV	RLWE	IND-CPA; noise budget mod t
BFV	RLWE	IND-CPA; noise budget under Δ
CKKS	RLWE	IND-CPA; approximate decryption
Regev / dual-Regev	plain LWE	the pedagogical starting point
GSW	plain LWE	approximate eigenvector encryption

Every scheme built on these problems carries an error in its ciphertexts. Three facts govern it:

1. **Addition adds errors** — sums are essentially free.
2. **Multiplication multiplies them** — the error grows exponentially in the multiplicative *depth*.
3. **Decryption works while the error stays below a fixed fraction of q .**

Because β bounds every fresh error, all three are arithmetic with the norm inequalities of Lecture 3.

Lecture 7 applies this to Regev; the BFV lecture turns it into a full noise calculus that BGV, CKKS and ML-KEM reuse.

1. Four distributions do the work: uniform on R_q , which absorbs anything added to it; ternary R_2 for keys; $D_{\mathbb{Z},8}$ with $(\mu, \sigma, \beta) = (0, 3.1915, 19)$ for errors; CBD_η for ML-KEM. It is the *bound* $\beta = \lfloor 6\sigma \rfloor$, not the distribution, that makes every later norm argument finite.
2. $\text{LWE} \rightarrow \text{RLWE} \rightarrow \text{MLWE}$ trades structure for size; MSIS is the short-kernel problem signatures need.
3. Addition adds errors, multiplication multiplies them, and decryption is correct while the error stays below a fraction of q — made precise in the scheme lectures.

Noise growth and correctness are taught in the BGV lecture, the BGV failure mode in the BGV lecture, the message layouts in the CKKS lecture, and the failure probability in the ML-KEM lecture.

Lecture 7 — Lattices, LWE, and Regev's Cryptosystem

This lecture fixed the distributions and the hardness assumptions.

The next lecture spends them on the first lattice cryptosystem: Regev's scheme, whose correctness is the first application of the noise facts previewed at the end of this lecture.